# Molecular basis of Inheritance (DPP)

## DPP - 1

**1.** From following example DNA, tRNA, rRNA, ribose, Deoxyribose and Uridine: How many are nucleic acid
1) Four
2) Three
3) Five
4) Six

**2.** Mark the incorrect statement-
1) Both sugar and nitrogenous bases are heterocyclic
2) Nucleotide and nucleoside have Nglycosidic bond
3) Polynucleotide chain lack phosphodiester bond
4) H-bond present between nitrogenous bases

**3.** Phosphoester bond present in
1) Uracil mono phosphate
2) Adenosine
3) Guanosine mono phosphate
4) Thymine

**4.** Difference in cytidine and deoxycytidine is:
1) Type of Bond between sugar and nitrogenous base
2) Type of Sugar
3) Phosphate
4) Nitrogenous base

**5.** One of the hallmarks of DNA double helix model was ……. between the two strands of polynucleotide chains
1) H-bond
2) Base pairing
3) Phosphoester bond
4) Antiparallel polarity

**6.** Mark the incorrectly matched:
1) ϕx174 - 5386 nucleotides
2) Bacteriophage lambda -48502 base pairs (bp)
3) Escherichia coli - $4.6 \times 10^6$ bp
4) Haploid content of human DNA - $6.6 \times 10^9$ bp

**7.** 'Nuclein' name was proposed by
1) Watson and Crick
2) Friedrich Meischer
3) Maurice Wilkins
4) Rosalind Franklin

**8.** According to Erwin Chargaff, ratios between purine and pyrimidine is
1) Constant and unequal
2) Inconstant and equal
3) Constant and equal
4) Inconstant and unequal

**9.** Who proposed the central dogma of molecular biology

1) Fransis Crick
2) James Watson
3) Mendal
4) Friedrich Miescher

**10.** Distance between two consecutive nucleotides of one polynucleotide chain in DNA is
1) 2 nm
2) 0.34 nm
3) 3.4 nm
4) None

**11.** Histones are positively charged in nature because of
1) Lysine and arginine
2) Lysine and Glycine
3) Lysine and isoleucine
4) Lysine and aspartate

**12.** Heterochromatin is a …….. chromatin
1) Densely packed and light stained
2) Densely packed and dark stained
3) Loosely packed and dark stained
4) Loosely packed and light stained

**13.** Statement-1: NHC proteins are a set of proteins used for packaging of chromatin
Statement-2: Chromatin is the coiled structure of nucleosome and linker DNA.
1) Both correct
2) Only Statement 1 is correct
3) Only Statement 2 is correct
4) Both incorrect

**14.** According to central dogma information flows from
1) DNA→ RNA → Protein
2) DNA → Protein → RNA
3) RNA→ DNA → Protein
4) Protein → RNA → DNA

**15.** If a DNA of 544 nm length, if % of Guanine is 40%, calculate the number of hydrogen bonds in a DNA molecule.
1) 4680
2) 4480
3) 5480
4) 5680

## DPP – 2

**16.** What was unique in Griffith experiments?
1) DNA: was found to be genetic material
2) RNA was found to be genetic material
3) Something from dead organisms could change the living cells
4) Viruses can live in bacteria.

**17.** Isotopes used by Hershey and Chase were
1) $^{32}P$ and $^{35}S$
2) $^{35}P$ and $^{32}S$
3) $^{34}P$ and $^{31}S$

4) $^{30}P$ and $^{32}S$

**18.** Hershey and Chase concluded that viral infecting agent in their experiment was

1) Protein
2) DNA
3) RNA
4) Both (2) and (3)

**19.** In the Griffith experiment, what would be the effect of following conditions on mice?

| | Form of Pneumococcus Injected | Effect on mice |
|---|---|---|
| i | live R- strain | A |
| ii | live S-strain | B |
| iii | Heat - killed S- strain | C |
| iv | Heat - killed S- strain + live R- strain | D |

Choose the correct option for effect on mice.

1) A- survived, B- died, C- died, Dsurvived
2) A- survived, B- died, C- survived, Ddied
3) A- died, B- survived, C- survived, Ddied
4) A- died, B- survived, C- died, D-Died

**20.** Hershey and Chase used 35S and 32P to prove that DNA is the genetic material. Their experiments proved that DNA is genetic material because

1) Progeny viruses retained $^{32}P$ but not $^{35}S$
2) Retention of $^{32}P$ in progeny viruses indicated that DNA was passed on
3) Loss of $^{35}S$ in progeny viruses indicated that proteins were not passed on
4) All of the above

**21.** Match the following

| | | | |
|---|---|---|---|
| A | Protease | i | Digestion of RNA |
| B | DNAse | ii | Breakdown of protein |
| C | RNAse | iii | DNA digestion |

1) A-i, B-ii, C-iii
2) A-ii, B-iii, C-i
3) A-iii, B-ii, C-i
4) A-i, B-iii, C-ii

**22.** Find out the correct sequence of steps with respect to the Hershey and Chase experiment?

1) Blending → infection → centrifugation
2) Infection → blending → centrifugation
3) Centrifugation → blending →infection
4) Infection → centrifugation → Blending

**23.** Which one of the following causes RNA to be easily degradable

1) Presence of OH at 2' site.
2) Presence of H at 2' site
3) Presence of OH at 3' site
4) Presence of H at 3' site

**24.** Who experimentally proved that DNA is semiconservative.

1) Mathew Meselson
2) Franklin Stahl
3) Both (a) and (b)
4) Watson and Crick

**25.** One of the characters of genetic material of having stability is proved by

1) Meselson

2) Griffith
3) Meischer
4) Crick

**26.** During Meselson and Stahl's experiments heavy DNA was distinguished from normal DNA by centrifugation in
1) CsOH gradient
2) $^{14}NH_4Cl$ gradient
3) $^{15}NH_4C1$
4) CsCl gradient

**27.** Radioactive isotope used by Taylor in his experiment was
1) Iron
2) Titanium
3) Thymidine
4) Uracil

**28.** Viruses having RNA genome and having shorter lifespan, mutate and evolve faster because
1) RNA is unstable and mutates at faster rate
2) RNA is stable and mutates at faster rate
3) RNA is stable and mutates at slower rate
4) RNA is unstable and mutates at a slower rate.

**29.** Stability of genetic material can be explained as
1) Genetic material does not change with every stage of life cycle.
2) DNA strands came together in appropriate condition after separation by heating.
3) DNA strands being complementary.
4) All of them.

**30.** Statement-1: DNA is considered to be better genetic material than RNA for most organisms.
Statement-2: Mutation rate of RNA is slower than DNA.
1) Both correct
2) Only Statement-1 is correct
3) Only Statement-2 is correct
4) Both incorrect.

## DPP-03

**31.** During DNA replication, DNA do not separate to its entire length because
1) This mechanism does not require energy
2) Ligase is unable to separate DNA segments.
3) It is very high energy required process
4) None of them

**32.** During DNA replication, okazaki fragments are used to elongate
1) The leading strand towards replication fork
2) The lagging strand towards replication fork
3) The leading strand away from replication fork
4) The lagging strand away from replication fork

**33.** The scheme of semiconservative DNA explains
1) Both strands act as template for synthesis of new DNA
2) Both strands are complementary
3) Both strands contain one strand from parental DNA
4) All of them

**34.** Replication of DNA occurs in
1) Bidirectional on both strands
2) Unidirectional on both strand
3) Bidirectional on single strand
4) None of them

**35.** Select the correct match of enzyme with its related function
1) DNA polymerase - synthesis of DNA strand
2) RNA polymerase - synthesis of DNA
3) Ligase - uncoiling of DNA
4) DNA polymerase - joining two DNA segment together

**36.** Light DNA in Meselson and Stahl experiment was
1) DNA obtained with 15-N in both strands
2) DNA obtained with 14-N in both strands
3) DNA obtained with more 14-N and few 15-N in both strands
4) DNA obtained with more 15-N and few 14-N in both strands

**37.** Experimental proof of DNA being semiconservative was given in year-
1) 1958
2) 1957
3) 1955
4) 1953

**38.** The position of promoter and terminator w.r.t coding strand is ___ and _____ respectively
1) 5'end downstream; 3' end upstream
2) 5'end upstream; 3' end downstream
3) 3'end downstream; 5' end upstream
4) 3' end upstream; 5' end downstream

**39.** Which of the following make use of DNA as a template to synthesize RNA –
1) DNA dependent RNA polymerase
2) DNA polymerase
3) Reverse transcriptase
4) RNA dependent DNA polymerase

**40.** Mark the incorrect pair:
1) Promoter - Binding site for RNA polymerase
2) Terminator - Define the end of transcription process
3) Cistron- Segment of RNA coding for a polypeptide
4) Exons - Expressed sequences

**41.** The factors that are responsible for the initiation and termination 'of transcription in eukaryotes are
1) $\sigma$ and $\rho$, respectively
2) $\rho$ and $\gamma$, respectively
3) $\alpha$ and ß, respectively
4) $\gamma$ and $\sigma$, respectively

**42.** In splicing, the sequences which are kept and those which are removed, respectively, are
1) Exons and introns
2) Introns and exons
3) Exons and cistrons
4) Introns and cistrons

**43.** What will happen if both the strands of DNA are copied during transcription?

1) Two RNA will be produced complementary to each other.
2) Double stranded RNA will be formed
3) No translation
4) All of the above

**44.** What will be the sequence of mRNA produced by the following stretch of DNA?
3'- TACGTACGTACGTAC-5' Template strand
5'- ATGCATGCATGCATG-3' Coding strand
1) 3'- UACGUACGUACGUAC-5'
2) 5'- UACGUACGUACGUAC-3'
3) 5'- AUGCAUGCAUGCAUG- 3'
4) 3'- AUGCAUGCAUGCAUG-5'

**45.** If the coding strand has the sequence 5'- ATCGATCG-3' then find out the sequence of non-coding strand.
1) 3'- TAGCTAGC-5'
2) 5'- TAGCTAGC-3'
3) 3'- UAGCUAGC-5'
4) 5'- UAGCUAGC-3'

**46.** With regards to mature mRNA in eukaryotes
1) Exons and introns appear in the mature RNA
2) Introns appear, but exons do not appear in the mature RNA
3) Exons appear, but introns do not appear in the mature RNA
4) Both exons and introns do not appear in mature RNA

**47.** Which is present at 5'end of eukaryotic mRNA?
1) Poly-A tail
2) Modified C at 5'
3) 7 mG
4) Poly-C

**48.** Match the following :-

| Column I | | Column II | |
|---|---|---|---|
| A | rRNA | 1 | Transfer of genetic material |
| B | mRNA | 2 | Nucleolar organizing region |
| C | rRNA | 3 | Transfer of amino acid from cytoplasm to ribosome |

1) A-1, B-2, C-3
2) A-2, B-3, C-1
3) A-3, B-1, C-2
4) A-2, B-1, C-3

**49.** Statement-1: tRNA is usually smaller than mRNA.
Statement-2: mRNA contains anticodons, but tRNA contains codons.
1) Both statements are correct.
2) Only statement 1 is correct.
3) Only statement 2 is correct.
4) Both statements are incorrect.

**50.** Which one of the following also act as a catalyst in a bacterial cell
1) 23S rRNA
2) 5S rRNA
3) snRNA
4) hnRNA

# DPP-04

**51.** Mark the correct combination of salient features of genetic code.
1) Universal, non-ambiguous, overlapping
2) Degenerate, overlapping, commaless
3) Universal, ambiguous, degenerate
4) Degenerate, non-overlapping, nonambiguous

**52.** Genetic code
1) Is a relationship between sequence of DNA or mRNA to polypeptide
2) Triplet base on mRNA
3) Determines the sequence of amino acid in polypeptide
4) All of the above

**53.** In order to code for all the 20 amino acids, the code should be made up of three nucleotides. This statement was suggested by
1) Har Gobind Khorana
2) George Gamow
3) Marshall Nirenberg
4) Servo Ochoa

**54.** Mutation can describe about-
1) Relationship between genes and DNA
2) Effects of rearrangement of DNA segment
3) Change in base pairs
4) All of them

**55.** Sickle- cell anaemia is a classic example of point mutation in which valine amino acid comes in place of
1) Glutamate
2) Tryptophane
3) Alanine
4) Guanine

**56.** _____ is the codon which has dual function.
1) UAG
2) UUU
3) AUG
4) AAA

**57.** Imagine an error occurring during DNA replication in a cell, so that where there is supposed to be A in one of the genes there is à C instead. What effect will this probably have on the cell?
1) The amino acid sequence will be completely changed.
2) An amino acid will be missing.
3) An incorrect amino acid will appear.
4) An additional amino acid will appear.

**58.** Glycine and alanine are code by how many codons respectively
1) 2 and 4
2) 4 and 6
3) 4 and 4
4) 6 and 2

**59.** In the protein synthesis, tRNA carrying the amino acid enters in which state in the ribosome?
1) Charged with energy form
2) Uncharged form

3) In previously formed peptide bond form
4) None of them

**60.** UTRs present on mRNA refers to
1) Untranscribed regions at both 5' end and 3' end
2) Untranslated regions at 5' end
3) Untranslated regions at both 5' end and 3' end
4) Untranslated regions at 3' end

**61.** Which of the following process occur during charging or aminoacylation of tRNA
1) Activation of amino acids in the presence of ATP
2) Linking of amino acids to their cognate tRNA
3) Both (1) and (2)
4) None of the above

**62.** Which of the following steps of translation does not have an energy requirement.
1) Ribosome in subunit state
2) Peptide bond formation
3) Charging of tRNA
4) Aminoacylation of tRNA

**63.** Termination of protein synthesis or translation requires
1) Both stop signal and starting codon
2) Both starting codon and release factor
3) Both release factor and stop codon
4) GUG and AUG codon

**64.** Ribozyme is catalyst in bacteria. it is …….. .acting as catalyst.
1) 23S rRNA
2) 5S rRNA
3) 28S rRNA
4) None of them

**65.** Statement-1: binding of mRNA to the ribosome is loose and reversible.
Statement-2: the ribosome, which is composed of tRNA and proteins, consists of a small and large subunit.
1) Both statements are correct.
2) Only statement 1 is correct.
3) Only statement 2 is correct.
4) Both statements are incorrect.

**66.** Which of the following is not produced by E. coli in the lactose medium?
1) β -galactose
2) Transacetylase
3) Lactose dehydrogenase
4) Permease

**67.** Regulator Adda24/7 gene controls chemical synthesis (operon system) by inhibiting
1) Substrate enzyme action
2) Transcription of mRNA
3) Passage of mRNA
4) Enzymes

**68.** Mark the correct one
1) i gene – repressor
2) z gene – permease

3) a gene - β -galactosidase
4) y gene - transacetylase

**69.** Mark the incorrect option with respect to lac operon
1) It is under positive as well as negative control
2) It controls catabolic pathway
3) It shows feedback repression
4) It was discovered by Jacob and monad

**70.** Sequencing the whole genome of an organism into coding and non-coding sequence is termed as
1) Sequence determination
2) Sequence detection
3) Sequence annotation
4) Sequence completion

## DPP-05

**71.** VNTR means
1) Variable number of tandem repeats
2) Various number of tandem repeats
3) Variable number of transcript repeats
4) Variable number of transcribed repeats.

**72.** Satellite DNA is
1) Non coding DNA
2) Repetitive DNA
3) Highly polymorphic
4) All of them

**73.** S-I: DNA polymorphism is the basis of DNA fingerprinting
S-II: DNA from every tissue of an individual shows the same degree of polymorphism.
1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II is correct
4) Both S-I and S-II is incorrect

**74.** Which of the following statement is incorrect from given below
1) DNA is classified on the basis of base composition present in it.
2) Satellite DNA does not code for any proteins.
3) DNA from every tissue from an individual show different degree of polymorphism
4) Both b and c

**75.** Technique of DNA fragment transferring onto nitrocellulose membrane is called
1) Southern blotting
2) Northern blotting
3) Western blotting
4) Eastern blotting

**76.** Attachment of radioactive probe with DNA fragments is known as
1) Blotting
2) Hybridisation
3) Electrophoresis
4) Isolation

**77.** For DNA fingerprinting isolation of DNA is done from which source cell?
1) Hair cell
2) Skin cell
3) Blood
4) All of them

**78.** Allelic sequence variation has been described as a DNA polymorphism if-
1) An allele occurs in human population at a locus with a frequency greater than 0.1
2) An allele occurs in human population at a locus with a frequency greater than 0.001
3) An allele occurs in human population at a locus with a frequency greater than 0.01
4) An allele occurs in human population at a locus with a frequency greater than 1.

**79.** Choose the correct one with respect to DNA fingerprinting application.
1) In forensic science
2) In determining the genetic diversity
3) In determining the population diversity
4) All of them

**80.** Assertion: The sensitivity of DNA fingerprinting can be increased by using polymerase chain reaction.
Reason: DNA fingerprinting detect the inheritable polymorphism from parents to children.
1) If assertion and reason is true and reason is correct explanation of assertion.
2) If assertion and reason is true and reason is not correct explanation of assertion.
3) If assertion is true and reason is false.
4) If both are false.

**81.** Polymorphism means
1) Variation at tissue level
2) Variation at species level
3) Variation at cellular level
4) Variation at genetic level

**82.** Steps of DNA southern blotting are as follows:
I. Digestion II. Hybridization III. Separation IV. Transfer of DNA
V. Isolation of DNA VI. Hybrid DNA detection
Arrange them in correct manner
1) I, II, III, IV, V, VI
2) V, I, III, IV, II, VI
3) V, II, I, III, VI, IV
4) III, II, I, IV, V, VI

**83.** The DNA fingerprinting technique was proposed by
1) Jacob and Monod
2) Fredrich Sanger
3) Har Gobind Khurana
4) Alac Jeffreys

**84.** Statement-1: Satellite DNA gives out bulk peaks in gel electrophoresis.
Statement-2: Satellite DNA is of two types: mini or micro.
1) Both statements are correct.
2) Only statement 1 is correct.
3) Only statement 2 is correct.
4) Both statements are incorrect.

**85.** Application of HGP include
1) Forensic science

2) Genetic biodiversity
3) Evolutionary biology
4) All of them

Answer Key

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **1)** | 2 | **16)** | 3 | **31)** | 3 | **46)** | 3 | **61)** | 3 | **76)** | 2 |
| **2)** | 3 | **17)** | 1 | **32)** | 4 | **47)** | 3 | **62)** | 1 | **77)** | 4 |
| **3)** | 3 | **18)** | 2 | **33)** | 4 | **48)** | 3 | **63)** | 3 | **78)** | 3 |
| **4)** | 2 | **19)** | 2 | **34)** | 1 | **49)** | 2 | **64)** | 1 | **79)** | 4 |
| **5)** | 2 | **20)** | 2 | **35)** | 1 | **50)** | 1 | **65)** | 2 | **80)** | 1 |
| **6)** | 4 | **21)** | 2 | **36)** | 2 | **51)** | 4 | **66)** | 3 | **81)** | 4 |
| **7)** | 2 | **22)** | 2 | **37)** | 1 | **52)** | 4 | **67)** | 2 | **82)** | 2 |
| **8)** | 3 | **23)** | 1 | **38)** | 2 | **53)** | 2 | **68)** | 1 | **83)** | 4 |
| **9)** | 1 | **24)** | 3 | **39)** | 1 | **54)** | 4 | **69)** | 3 | **84)** | 3 |
| **10)** | 2 | **25)** | 2 | **40)** | 3 | **55)** | 1 | **70)** | 3 | **85)** | 4 |
| **11)** | 1 | **26)** | 4 | **41)** | 1 | **56)** | 3 | **71)** | 1 | | |
| **12)** | 2 | **27)** | 3 | **42)** | 1 | **57)** | 3 | **72)** | 4 | | |
| **13)** | 1 | **28)** | 1 | **43)** | 4 | **58)** | 3 | **73)** | 3 | | |
| **14)** | 1 | **29)** | 4 | **44)** | 3 | **59)** | 1 | **74)** | 3 | | |
| **15)** | 2 | **30)** | 2 | **45)** | 1 | **60)** | 3 | **75)** | 1 | | |